

Ex.No. 11: Implementation of IoT-based Smart Agriculture using MATLAB

AIM: To design and simulate an IoT-based smart agriculture system using MATLAB by

monitoring soil moisture levels, controlling water pump status (ON/OFF), and evaluating

irrigation efficiency for optimized water resource management.

Objective:

1. To monitor the Soil Moisture (%) level in agricultural fields using IoT-based sensing

simulation in MATLAB.

2. To control and display the Water Pump Status (ON/OFF) based on threshold soil

moisture conditions.

3. To evaluate the Irrigation Efficiency (%) by analyzing water usage effectiveness and

moisture maintenance level.

Procedure

1. Define soil moisture range (0–100%) and set an irrigation threshold value (e.g., 40%).

2. Generate or input soil moisture data to simulate field conditions.

3. Compare soil moisture with the threshold value to determine pump status (ON if low

moisture, OFF if sufficient moisture).

4. Simulate water pumping action based on control logic.

5. Calculate irrigation efficiency based on moisture improvement and water usage.

6. Display soil moisture level, pump status, and efficiency results.

7. Plot graphs to analyze soil moisture variation and irrigation performance.

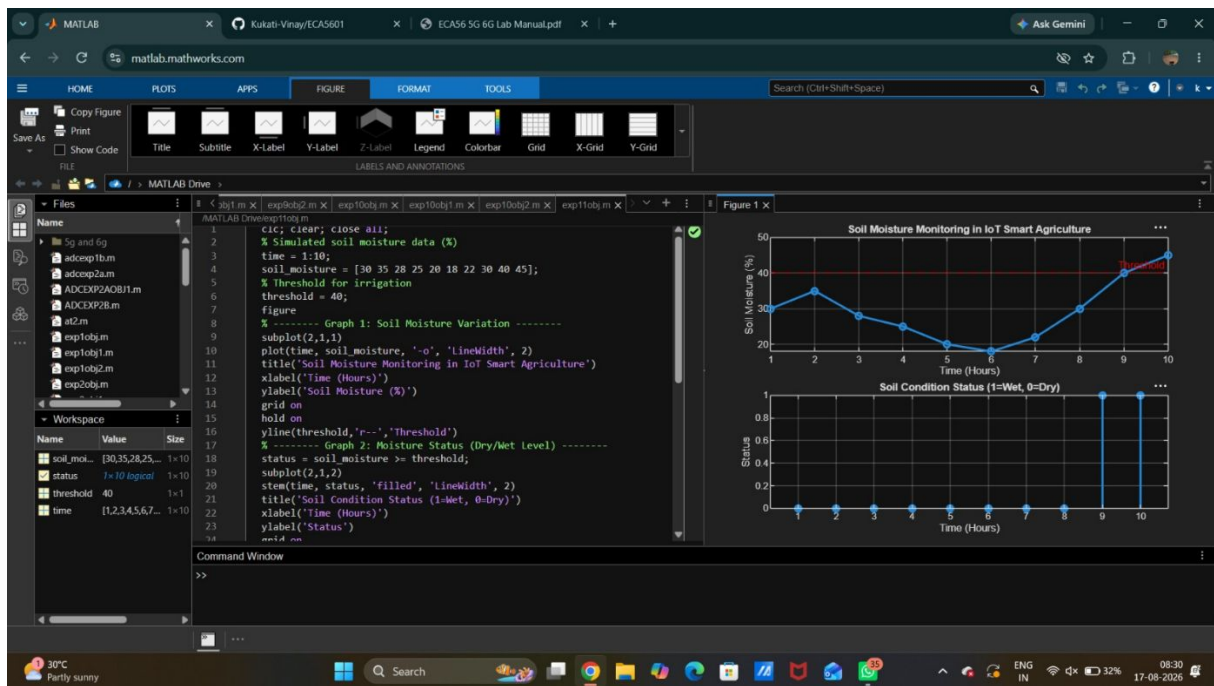
8. Evaluate system behavior for smart agriculture decision-making.

Objective1

Program:

```
clc; clear; close all;
% Simulated soil moisture data (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold for irrigation
threshold = 40;
figure
% ----- Graph 1: Soil Moisture Variation -----
subplot(2,1,1)
plot(time, soil_moisture, '-o', 'LineWidth', 2)
...yline(threshold,'r--','Threshold')
% ----- Graph 2: Moisture Status (Dry/Wet Level) -----
status = soil_moisture >= threshold;
subplot(2,1,2)
stem(time, status, 'filled', 'LineWidth', 2)
title('Soil Condition Status (1=Wet, 0=Dry)')
xlabel('Time (Hours)')
ylabel('Status')
grid on
```

output:



Objective2:

Program:

```
clc; clear; close all;
```

```
% Simulated soil moisture (%)
```

```
time = 1:10;
```

```
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
```

```
% Threshold level
```

```
threshold = 40;
```

```
% Pump control logic (1 = ON, 0 = OFF)
```

```
pump_status = soil_moisture < threshold;
```

```
figure
```

```
% ----- Graph 1: Soil Moisture vs Pump Status -----
```

```
...hold on
```

```
ylabel(threshold,'r--','Threshold')
```

```
% ----- Graph 2: Pump Status (ON/OFF) -----
```

```
subplot(2,1,2)
```

```
stem(time, pump_status, 'filled','LineWidth',2)
```

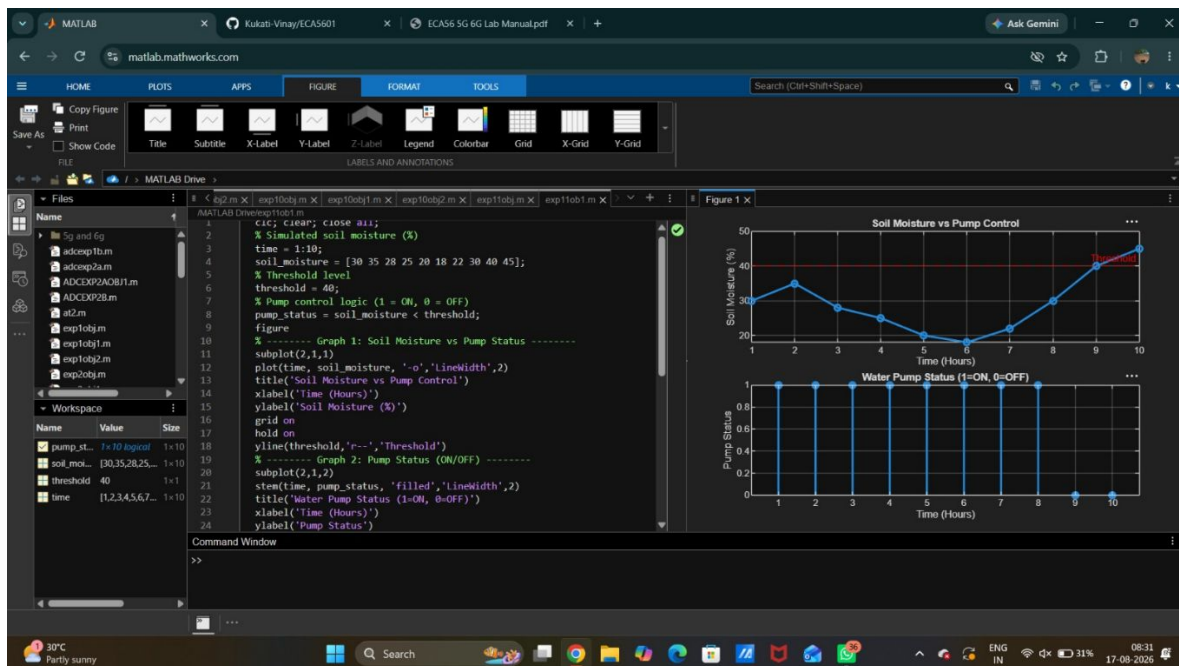
title('Water Pump Status (1=ON, 0=OFF)')

xlabel('Time (Hours)')

ylabel('Pump Status')

grid on

output:



Objective3:

Program:

```
clc; clear; close all;
```

```
% Simulated soil moisture data (%)
```

```
time = 1:10;
```

```
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
```

```
% Threshold level
```

```
threshold = 40;
```

```
% Irrigation efficiency calculation (%)
```

```
efficiency = (soil_moisture ./ threshold) * 100;
```

```
efficiency(efficiency > 100) = 100; % limit to 100%
```

```
figure
```

...grid on

hold on

ylines(threshold,'r--','Threshold')

% ----- Graph 2: Irrigation Efficiency -----

subplot(2,1,2)

plot(time, efficiency, '-s','LineWidth',2)

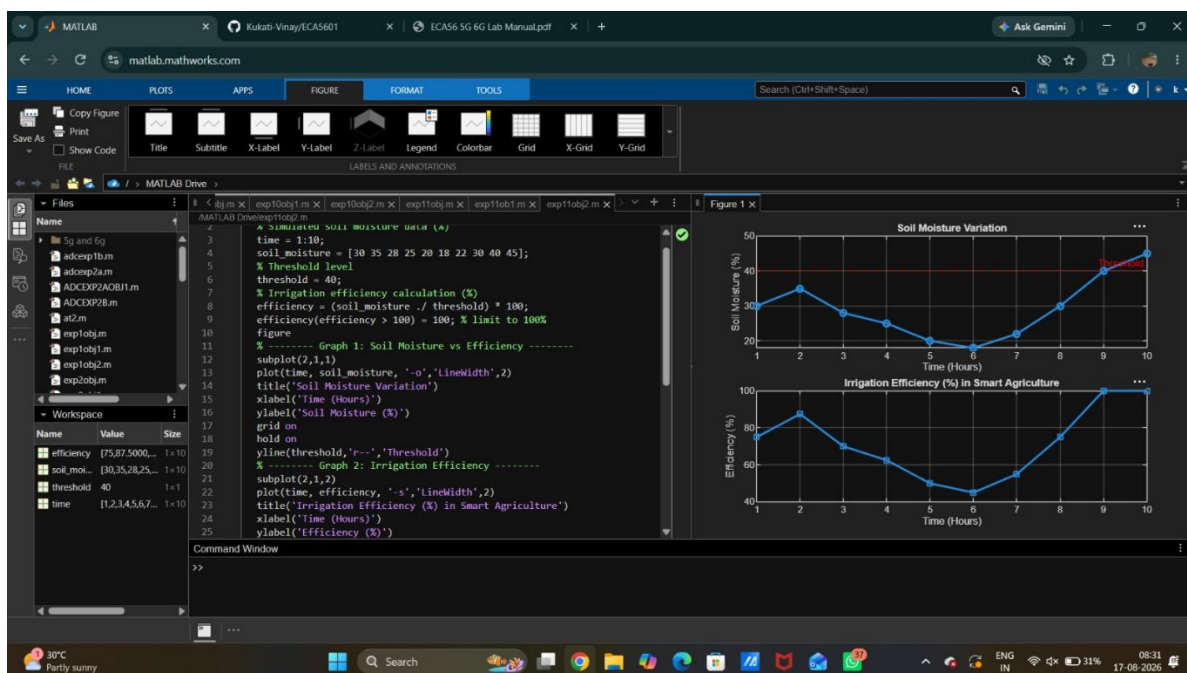
title('Irrigation Efficiency (%) in Smart Agriculture')

xlabel('Time (Hours)')

ylabel('Efficiency (%)')

grid on

output:



Result:

1. The soil moisture levels were successfully monitored using MATLAB simulation, showing variation over time and identifying dry and wet conditions based on threshold values.

2. The water pump control system was effectively implemented, where the pump automatically switched ON when soil moisture dropped below the threshold and remained OFF when moisture was sufficient.
3. The irrigation efficiency was successfully calculated and analyzed, showing that the system optimizes water usage by maintaining soil moisture close to the desired level.